Name: Darshika Singh
Sem:5 Div:T2
Batch:T22
Roll no:2401114
Date : 15/07/2026

Experiment 01

Aim:Version control operations using Git.

This experiment maps LO2

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory: Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

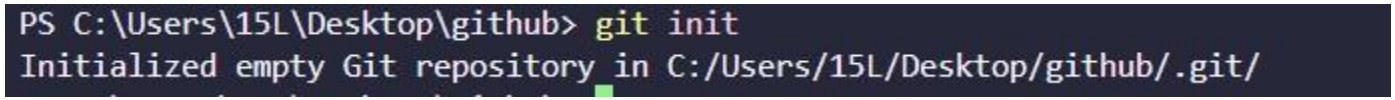
Command	Syntax	Purpose
Initialize Repository	git init	Creates a new Git repository.
Version Check	git version--	Checks the git version
Check Status	git status	Shows modified, staged, and untracked files.
Master Branch	git branch master -m main	Changes Master branch name to main
Stage All Files	git add .	Adds all changes to the staging area.
Stage One File	git add python.py	Adds only python.py to staging.
Commit Changes	git commit -m "message"	Saves staged changes with a message.
View Commit History	git log	Shows complete commit history.
One-line History	git log --oneline	Shows compact commit history.
Push to GitHub	git push origin main	Uploads local commits to the remote main branch.

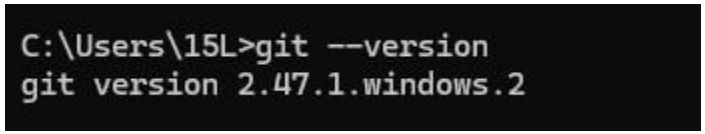
Clone Repository	git clone <url>	Copies a GitHub repository to your computer.
List Branches	git branch	Displays all local branches.

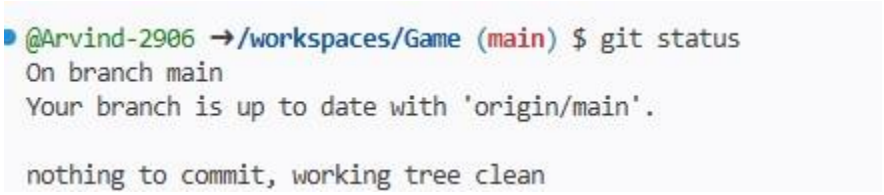
Command	Syntax	Purpose
Switch Branch	git checkout main	Switches to the main branch.
Create & Switch	git checkout -b arvind	Creates and switches to a new branch.
Push Branch	git push origin arvind	Pushes the arvind branch to GitHub.
Pull Changes	git pull origin main	Fetches and merges changes from the remote main.
Fetch Only	git fetch origin	Downloads remote updates without merging them.
Merge Branch	git merge arvind	Merges arvind into the current branch.
Restore File	git restore python.py	Discards uncommitted changes in python.py.
Restore All Files	git restore .	Discards all uncommitted changes.
Unstage File	git restore --staged python.py	Removes python.py from the staging area.
Soft Reset	git reset --soft HEAD~1	Removes the last commit but keeps changes staged.
Hard Reset	git reset --hard HEAD~1	Removes the last commit and discards all changes.
Reset to Remote	git reset --hard origin/main	Makes your local branch exactly match origin/main.
Show Differences	git diff	Displays unstaged changes.
Show Staged Differences	git diff --staged	Displays staged changes.
Show Remote	git remote -v	Lists configured remote repositories.

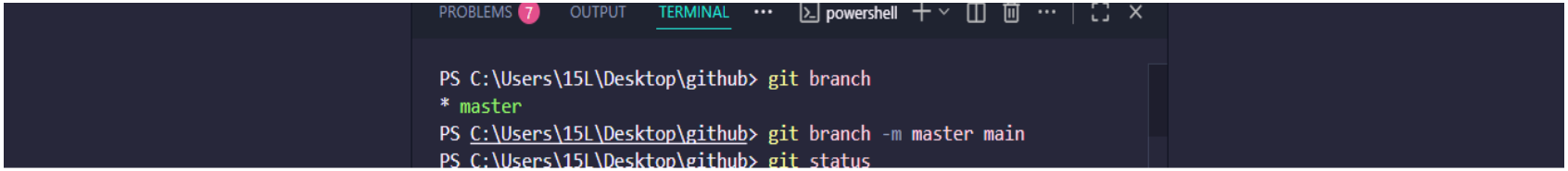
Add Remote	git remote add origin <url>	Connects a local repository to GitHub.
Delete Branch	git branch -d arvind	Deletes the local arvind branch.
Delete Remote Branch	git push origin --delete arvind	Deletes the remote arvind branch.
Force Push Safely	git push --force-with-lease origin main	Rewrites the remote branch after local history changes, with a safety check.
Set Username	git config --global user.name "Arvind-9206"	Sets your Git username globally for all repositories.
Set Email	git config --global user.email	Sets your Git email globally for all repositories.
Command	**Syntax**	**Purpose**
	"arvindpatil.9206@gmail.com"	

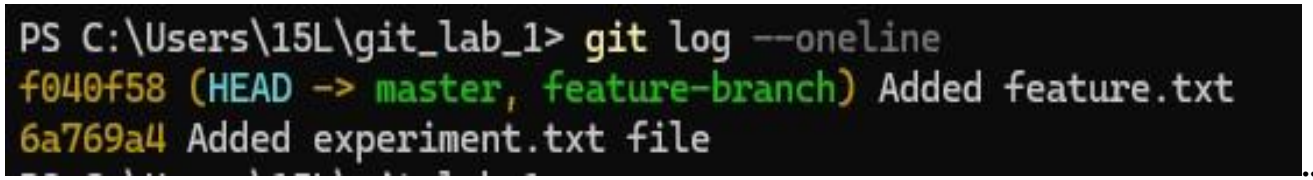
Output



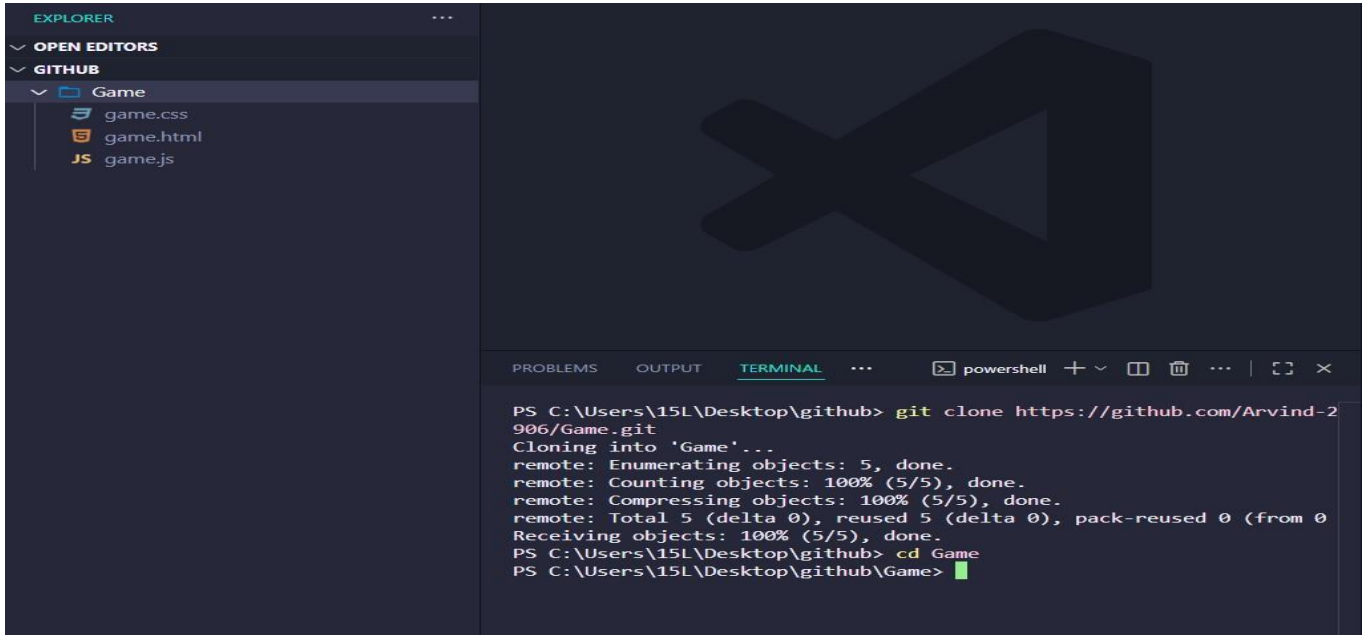





PROBLEMS 7
OUTPUT
TERMINAL
powershell
PS C:\Users\15L\Desktop\github> git branch
* master
PS C:\Users\15L\Desktop\github> git branch -m master main
PS C:\Users\15L\Desktop\github> git status


PS C:\Users\15L\git_lab_1> git log --oneline
f040f58 (HEAD -> master, feature-branch) Added feature.txt
6a769a4 Added experiment.txt file

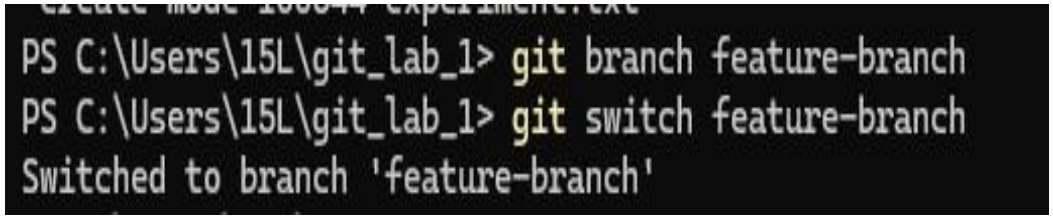

@Arvind-2906 →/workspaces/Game (main) $ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 294 bytes | 294.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Arvind-2906/Game
462227d..4931fc0 main -> main

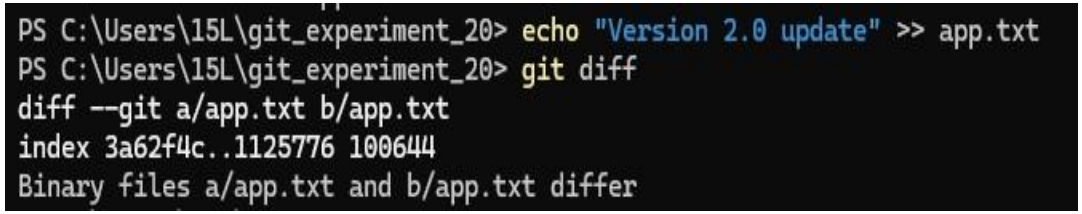

EXPLORER
OPEN EDITORS
GITHUB
Game
game.css
game.html
game.js
PROBLEMS
OUTPUT
TERMINAL
powershell
PS C:\Users\15L\Desktop\github> git clone https://github.com/Arvind-2906/Game.git
Cloning into 'Game'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0
Receiving objects: 100% (5/5), done.
PS C:\Users\15L\Desktop\github> cd Game
PS C:\Users\15L\Desktop\github\Game>

.

.....

, ..


PS C:\Users\15L\git_lab_1> git branch feature-branch
PS C:\Users\15L\git_lab_1> git switch feature-branch
Switched to branch 'feature-branch'



CONCLUSION

The experiment demonstrated the use of essential Git version control commands for repository management, tracking changes, branch handling, and collaboration with GitHub. It provided practical knowledge of managing and maintaining software projects efficiently using Git.